

## **User Manual**

### **FIT AR Navigation App (FITARNA)**

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## **1. Introduction**

FITARNA is an augmented reality navigation application designed for use inside the Florida Tech library. The app's purpose is to help users navigate the library more easily while also providing an interactive experience through augmented reality visuals and informational pop-ups.

Using FITARNA, users can select different floors and destinations within the library and receive visual guidance to help them reach their chosen location. In addition to navigation, the app enhances the user experience by displaying informational AR pop-ups that share the library's history, fun facts, and other helpful details.

FITARNA is intended to make the library easier to explore for students, faculty, staff, and visitors. Whether a user wants to quickly find a destination or casually explore the building, the application provides a more engaging and intuitive way to navigate the space.

If additional help is needed after downloading or using the app, users may contact the FITARNA development team using the email addresses listed at the beginning of this manual.

## **2. Downloading the App**

## **2.1. Compatible Systems**

At launch, FITARNA is available only on Android devices. The app is primarily designed for Android smartphones, but it may also work on compatible Android tablets.

Because FITARNA uses augmented reality, the device must support the hardware and software features required for AR functionality. For the best experience, users should use a modern Android device with a working camera, motion tracking capability, and up-to-date system software.

Support for iPhone and other platforms may be considered in future versions of the application.

## **2.2. How to Install**

FITARNA is installed by scanning a provided QR code that links directly to the application's .apk download file.

To install the app:

- Open the camera app or a QR code scanner on your Android device.
- Scan the QR code provided for FITARNA.
- Tap the download link for the .apk file when it appears.
- Follow the on-screen prompts to download and install the application.

Some Android devices may block the installation of applications downloaded outside of the default app marketplace. If this happens, the device may prompt the user to allow installation from unknown or external sources. If prompted, the user should temporarily enable this permission in their device settings to complete the installation.

After the installation finishes, the FITARNA app icon should appear on the device. Open the application and grant any required permissions, such as camera access, so the augmented reality features function correctly. Without these permissions, the app could not function as intended.

### **2.3. Updates**

FITARNA may receive updates over time to improve performance, fix bugs, enhance navigation accuracy, and add new features.

When a new version of the application is released, users may be asked to install a newer .apk file via an updated QR code or download link. It is recommended that users install the latest available version to ensure the best performance and the most reliable user experience. Using an outdated version may result in reduced performance and missing improvements.

### **2.4. Getting Started**

Once FITARNA is installed/updated, the user can open the application from their device like any other mobile app. Users should already be physically inside the library when starting the app.

Once the user is in the library and the app is open, users can optionally take a moment to look around at their surroundings with their phone camera to help the app localize. Then users can choose between the two app modes from the apps home screen. Users who want to go directly to a room or area may select Navigation Mode, while users who want to experience the informational tour may select Tour Mode.

### **3. App Modes**

#### **3.1. Tour Mode**

Tour Mode allows users to explore the library freely while viewing augmented reality pop-ups placed throughout the environment. This mode can be activated by selecting the Tour Mode button on the FITARNA home screen.

When Tour Mode is active, users can move through the library at their own pace and view informational content connected to different areas of the building. These pop-ups may include historical details, trivia, or interesting facts about the library and its spaces.

Tour Mode is intended to create a more engaging and educational experience for users who want to learn more about the library rather than just visit a single destination. It is especially useful for first-time visitors, prospective students, and anyone interested in exploring the space interactively.

### **3.2. Navigation Mode**

Navigation Mode helps users travel directly to a selected destination within the library. This mode can be activated by selecting Navigation Mode from the FITARNA home screen.

Once Navigation Mode is opened, the user can choose a destination within the library. FITARNA will then guide the user using augmented reality navigation cues and visual guidance to help them reach that location.

This mode is useful for users who need help locating a specific room, service area, study space, or other point of interest inside the building. By offering real-time guided navigation, FITARNA reduces confusion and makes indoor wayfinding easier and more accessible.

## **4. App Pop-ups**

FITARNA includes in-app augmented reality pop-ups that appear as virtual information boxes within the user's view. These pop-ups are designed to provide useful and engaging information as users navigate the library. They are one of the main features of the app that distinguishes FITARNA from other navigation applications.

The pop-ups are intended to be easy to notice, quick to read, and visually clear. Their purpose is to share information about the library in a format that supports the user experience without becoming distracting. They appear in the augmented reality view to feel more connected to the user's surroundings.

Depending on the location and app mode, these pop-ups may display:

- Historical information about the library
- Fun facts or trivia
- General points of interest
- Helpful context related to the user's surroundings

These features help make FITARNA more than just a navigation tool. They also allow the app to serve as an interactive guide to the library environment.

## **5. Troubleshooting**

If FITARNA does not function as expected, there are a number of factors that could be at play.



### **5.1. Permissions**

If the user starts the FITARNA app and is presented with a black screen after selecting either mode, the likely cause is a permission issue. Augmented reality features depend on camera access, so denying this permission will prevent the app from displaying the user's view or any AR content.

To fix this issue, enable permissions in settings and restart the app.

### **5.2. Localizing**

If tracking or AR content appears inaccurate, the issue is likely a localization issue. The FITARNA app relies on the camera and saves scans of the library to localize. If the user's setting in the library differs from the one the dev team scanned during development, then it is likely that the app will be confused on user position. Additionally, if the user is moving too quickly, or isn't displaying their position and movement to the camera, localization is bound to be lost.

A way to aid localization is to point the camera at close up simple objects. Pillars and walls are great at relocalizing the user quickly. Make sure when moving around to keep the camera at a consistent height and point it in the direction of travel.

If localization problems persist after trying these steps, the user should restart the FITARNA application and try again in a well-lit area of the Evans Library.